Chapter 8
Collateral, Margin, and Funding

Several chapters have now deferred a question to this one. Chapter 2 showed margin calls transmitting stress across a market; Chapter 6 showed them threatening a hedged basis position; Chapter 7 showed the collateral agreement determining a derivative's discount curve. Each treated collateral as a means to some other end. This chapter treats it as the subject. The reason is a proposition that would have seemed eccentric twenty years ago and is now the organizing fact of derivatives risk management:

> A position can be perfectly hedged against price risk and still destroy the firm holding it, because the hedge demands cash on a schedule the hedged item does not supply.

This is not a hypothetical. In 2022 it happened three times in one year (in European energy, in LME nickel, and in UK pensions), to firms in unrelated businesses, none of which had misjudged the direction of any market. Sections 40.7, 40.8, and 40.12 tell all three episodes. The mechanism is always the same, and it lives entirely in the collateral arrangements that earlier chapters passed over.

Sections 8.1 to 8.3 cover the collateral agreement itself (the CSA, variation and initial margin, haircuts), Sections 8.4 and 8.5 the rules for uncleared trades and the optionality a CSA can carry, and Sections 8.6 to 8.8 what happens to collateral once posted (transformation, rehypothecation, segregation, disputes). Sections 8.9 and 8.10 work the proposition above through the 2022 energy and pension episodes and show how the resulting liquidity risk is measured and managed. Section 8.11 asks whether there is enough collateral in the system, and Section 8.12 says where all this leaves valuation, the starting point of Chapter 9.

8.1 THE CREDIT SUPPORT ANNEX

Derivatives between two parties are documented under an ISDA Master Agreement, which Section 1.5 described as providing close-out netting. Collateral is governed by the CSA, a separate document attached to it.

The CSA is where the economics of a trade are quietly decided. Its principal terms are:

- **Eligible collateral.** Which assets may be posted (cash in which currencies, which government bonds, occasionally corporate bonds or equities), and the haircut applied to each, the percentage by which its market value is marked down when it is counted (Section 8.3).
- *The interest rate paid on collateral.* Usually the overnight rate of the currency posted. Section 7.4 showed that this single line determines the discount curve for the whole trade.
- **Threshold.** An amount of exposure below which no collateral is required, in effect an unsecured credit line. The uncleared margin rules of Section 8.4 require a zero threshold on variation margin between the firms they cover, so a threshold now survives only on initial margin (the allowance of Section 8.4) and on legacy or uncovered CSAs.

- **Independent amount.** Collateral one party posts on top of whatever the mark-to-market requires, fixed in advance or set by formula. Before the uncleared margin rules it was how dealers took initial margin from hedge funds.
- **Rating triggers.** Credit-dependent terms: a threshold that falls, or an independent amount that rises, when a party is downgraded, so that a downgrade becomes a collateral call. Such clauses turned the 2008 downgrade of American International Group (AIG) into calls it could not meet (Section 40.9).
- **Minimum transfer amount.** A floor on the size of any single call, to avoid transferring trivial sums daily.
- *Frequency and timing.* How often the portfolio is valued and calls are made, how quickly a call must be met, and how disputes are handled.
- *Direction.* Whether both parties post (a two-way CSA) or only one (a one-way CSA).

These terms combine into a single number each day. The **credit support amount** is the collateral the receiver is entitled to hold:

$$\text{credit support amount} = \text{exposure} + \text{independent amount} - \text{threshold},$$

floored at zero, and the call is the difference between it and the collateral already held, valued after haircuts. Example 8.1 works one through.

Example 8.1 Computing a margin call

A dealer faces a corporate client under a CSA outside the uncleared margin rules. The client posts an independent amount of \$5 million and has a threshold of \$10 million; the minimum transfer amount is \$500,000, and calls are rounded up to the next \$100,000. The dealer holds collateral worth \$12.0 million after haircuts.

At today's valuation the portfolio is worth \$23.47 million to the dealer. The credit support amount is

$$23.47 + 5 - 10 = \$18.47\text{m},$$

so the shortfall is $18.47 - 12.0 = \$6.47$ million. That exceeds the minimum transfer amount, and rounding makes the call \$6.5 million. The dealer issues it on the valuation date and the client delivers within a business day or two, the settlement period the CSA sets. The dealer then holds \$18.5 million.

The next day the portfolio is worth \$23.2 million. The credit support amount falls to \$18.2 million and the dealer holds \$0.3 million more than it is entitled to, but that is below the minimum transfer amount, so nothing moves.

Even when every call is met, the dealer is collateralized only to within the threshold net of the independent amount: on the first day it held \$18.5 million against \$23.47 million of exposure. That gap of nearly \$5 million is the unsecured credit line Chapter 9 prices.

Two-way and one-way. In a one-way CSA the party that posts is traditionally the dealer, when the client is a sovereign, supranational, or large corporate with the bargaining power to insist. Since the post-crisis CVA charge and funding costs made such arrangements expensive for dealers, a number of sovereign debt offices have moved to two-way CSAs to obtain better pricing, but one-way arrangements remain common for sovereigns and supranationals, which the uncleared margin rules exempt.

One-way CSAs look like a commercial concession and are in fact a funding commitment. The dealer hedges the client trade with an offsetting trade that is itself collateralized. When the client trade moves in the dealer's favor, the hedge moves against it, so the dealer posts collateral

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on the hedge and receives nothing from the client; it therefore funds that collateral for the life of the trade, out of its own resources, at its own cost of funds. This is the origin of the funding valuation adjustment of Chapter 9, and it is why a trade that appears to be priced at the mid-market level can be substantially loss-making once the CSA is read.

Title transfer and security interest. Two legal forms exist. Under **title transfer**, collateral becomes the recipient's property outright, to be returned in equivalent form later; the recipient may use it freely, including by passing it on to a third party. Under a **security interest**, the poster retains ownership and the recipient holds a charge, a legal claim on the assets that it can enforce only if the poster defaults. The English-law CSA is a title transfer; the New York-law CSA creates a security interest, but its Paragraph 6(c) gives the recipient a right to use the collateral, including to rehypothecate it (Section 8.7), unless the parties disapply that right. The distinction is invisible until a default, at which point it determines whether the poster is an owner reclaiming property or an unsecured creditor in a queue.

8.2 VARIATION MARGIN AND INITIAL MARGIN

Two quite different things are both called margin, and conflating them is the most common error in this area.

Variation margin (VM) covers exposure that has *already arisen*. Each day the portfolio is revalued and the party for whom it has moved unfavorably transfers that amount. VM is a transfer of value that has already been lost; it does not make anyone worse off in economic terms, since the payer has suffered the loss whether or not any collateral changes hands.

Initial margin (IM) covers exposure that *might arise in the future*: specifically, the further move that could occur between a counterparty's last VM payment and the completion of close-out. It is posted at inception, recalculated daily by a model, and held, in a changing amount, until the trade matures. The distinction matters enormously for funding.

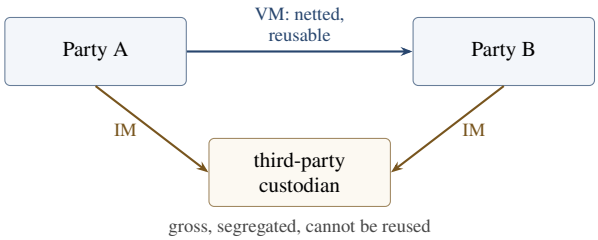


Figure 8.1: Variation margin passes between the parties; initial margin is posted gross to a custodian.

As Figure 8.1 shows, VM flows between the parties and is netted: A pays B or B pays A, never both. The recipient may use it, so it is not withdrawn from the system. IM is different: *both* parties post, *gross*, to a segregated account with a third-party custodian, and neither may touch it. The figure shows the bilateral arrangement under the uncleared margin rules; at a clearing house only the member posts initial margin. IM is therefore genuinely encumbered for the life of the trade, and it must be funded.

Example 8.2 Variation and initial margin on the same portfolio

A fund and a dealer hold a swap portfolio. At inception it is worth zero, and each posts initial margin of $25 million to a custodian. Neither can use the other's.

Rates then move and the portfolio is worth −$40 million to the fund. The fund pays $40 million

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of variation margin to the dealer, who may use it.

The two obligations are not comparable:

	Variation margin	Initial margin
Covers	loss already incurred	possible future move
Direction	one way, netted	both ways, gross
Amount	\$40m from fund	\$25m each
Reusable by recipient	yes	no
Economic cost to poster	none (value already lost)	funding spread, life of the trade

Posted IM is not idle in every sense: cash earns interest and bonds keep their yield. What it costs is the poster's funding spread over that return. At the 60 basis point spread Chapter 9 uses for this cost, the fund's IM costs it

$$\$25\text{m} \times 0.0060 = \$150{,}000 \text{ a year},$$

and the dealer bears the same on its own posting: a real cost of doing the trade that appears nowhere in the swap's valuation.

8.3 HAIRCUTS

Collateral is not counted at its full market value. A haircut (Section 4.1) discounts it, so that the collateral remains sufficient even if its own price falls before it can be liquidated.

If the haircut is h, then covering an exposure E requires collateral with market value

$$M = \frac{E}{1 - h} \tag{8.1}$$

Haircuts reflect the collateral's price volatility, its liquidity, and the time it would take to sell. Cash in the CSA's own currency carries none. Short-dated government bonds carry half a percent to two percent; long-dated government bonds more, because their prices move more; corporate bonds and equities considerably more. For firms that do not model their own, the uncleared margin rules of Section 8.4 prescribe a standardized schedule:

- cash, none;
- high-quality government debt, 0.5% up to one year to maturity, 2% from one to five years, and 4% beyond five;
- either of these in a currency other than the one the trade settles in, a further 8% currency add-on.

The currency add-on bears on the currency choice of Section 8.5. It is waived for cash variation margin in a major currency, where most of that option lives, but it applies to bonds and to initial margin posted in another currency. Example 8.3 shows what happens when a haircut widens.

Example 8.3 Haircuts and how they behave under stress

A fund must cover a £100 million exposure with gilts (UK government bonds) subject to a 4% haircut. From (8.1) it must deliver

$$\frac{100}{0.96} = £104.2 \text{ million of gilts}$$

Markets then become volatile and the haircut is raised to 10%. Covering the same £100 million now requires

$$\frac{100}{0.90} = £111.1 \text{ million}$$

an additional £6.9 million of gilts—on an exposure that has not changed at all.

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The example holds the exposure fixed; a real stress does not. When markets turn volatile enough to raise the haircut, three things happen together, and they compound. The exposure itself has probably grown, because the volatility that widened the haircut also moved the portfolio. The gilts have probably fallen in price, so the fund holds fewer pounds of collateral than it did. And the haircut has widened. The fund must find more assets, from a smaller pool, to cover a larger exposure. This is the same procyclicality Section 2.9 identified in initial margin models, operating through a second channel. A firm that stress-tests its margin requirements using *current* haircuts has tested the easy case.

8.4 UNCLEARED MARGIN RULES AND SIMM

Before the financial crisis, initial margin on bilateral derivatives was rare between dealers. Dealers did collect it from hedge funds, as independent amounts, but held it commingled with their own assets, which the failure of Lehman Brothers exposed. The post-crisis reforms of Section 2.11 pushed standardized products into clearing; for what remained bilateral, regulators imposed the uncleared margin rules (UMR), phased in over several years by size of counterparty and now applying broadly.

UMR requires variation margin on uncleared trades between covered firms, and initial margin as well once both groups hold more than €8 billion of uncleared derivatives notional ($8 billion in the United States), averaged over a prescribed window. Even then the parties may agree on a threshold of up to €50 million per counterparty group and post only the excess, to a segregated third-party custodian. The intent is that staying bilateral should not be a way to avoid posting collateral. Initial margin reached the largest dealers first and, in the final phase in September 2022, firms on the investing side such as asset managers and pension funds above the notional threshold. Some funds deliberately keep their uncleared notional below it so as never to post initial margin.

Calculating IM. Firms may use a model (which some regulators must approve) or a standardized schedule. Almost all use the model, because the schedule is punitive, and almost all use the same one: **ISDA SIMM**, an industry-standard methodology. The industry converged on one model deliberately, for a reason Section 8.8 explains. Adopting it is not automatic: in the European Union, the European Market Infrastructure Regulation as revised in 2024 brings in supervisory authorization of initial margin models, starting with the largest groups, and gives the European Banking Authority a central role in validating industry-wide models such as SIMM.

SIMM works from sensitivities rather than simulation. Each risk factor sensitivity is multiplied by a prescribed risk weight, and the weighted sensitivities are aggregated using prescribed correlations.

Example 8.4 A simplified SIMM calculation

A portfolio has interest rate sensitivity in two tenor buckets, the maturity ranges into which SIMM groups rate sensitivities:

Bucket	Sensitivity per bp	Risk weight (bp)	Weighted sensitivity
2-year	$80,000	60	$4.8m
10-year	$120,000	50	$6.0m

With a prescribed correlation of $\rho = 0.7$ between the two buckets, the margin is

$$\text{IM} = \sqrt{4.8^2 + 6.0^2 + 2 \times 0.7 \times 4.8 \times 6.0} = \sqrt{99.36} = \$9.97\text{m}$$

Compare this with the two figures that bracket it. With zero correlation the two charges would

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combine as $\sqrt{4.8^2 + 6.0^2}$ = \$7.68m. Simple addition of the weighted sensitivities gives \$10.8m, what would be required with no diversification recognized.

The diversification benefit here is modest, only 8% below the undiversified sum, because rate buckets are highly correlated. The benefit is larger where correlations are lower, as between interest rates and FX, which SIMM treats as separate risk classes within one product class. Across its four product classes (rates and FX, credit, equity, commodity) it recognizes none: their margins are simply added. Netting, too, stops at the edge of a relationship, since each counterparty margins only its own portfolio. This is why firms concentrate trades with fewer counterparties: splitting an offsetting book across two dealers means posting IM twice, on two portfolios each less diversified than the whole.

Cleared against bilateral

Because IM is model-driven, the same trade can require quite different amounts depending on where it is booked. Two features drive the gap.

The **margin period of risk** (the assumed time to close out a defaulted portfolio) is conventionally five days for cleared trades and ten for bilateral ones, on the reasoning that a CCP has a rehearsed auction process and a bilateral counterparty does not. Since margin scales roughly with the square root of the holding period, that alone makes bilateral IM about $\sqrt{10/5} = 1.41$ times larger.

Netting then compounds it. At a CCP a member's trades net across all its counterparties; bilaterally, each relationship nets only within itself. Example 8.5 puts numbers on both features.

Example 8.5 The same trade, cleared and bilateral

A fund executes a swap whose five-day 99% risk (the loss over five days that is exceeded with only 1% probability) is \$12 million.

Cleared. IM is \$12 million, posted once to the CCP. The dealer on the other side faces the CCP as well, but its leg offsets against the rest of its large cleared book and adds little to its own IM, so the trade's own cost is the fund's \$12 million.

Bilateral. The ten-day margin period raises it to roughly $12\sqrt{2}$ = \$17 million. Under UMR both parties post, gross and segregated. Other trades in the relationship have already used up its €50 million threshold, so each posts the full \$17 million and \$34 million is immobilized against the same economic risk.

Immobilized collateral is therefore nearly three times larger in the bilateral case. Without that offset (a dealer whose cleared book did not absorb the leg), the comparison would be \$24 million against \$34 million, and the gap would be the margin period's factor of 1.41 alone. Add the diversification point of Example 8.4 (the fund's other trades net at the CCP but not across separate bilateral relationships) and the difference widens further.

The rules were designed to make clearing cheaper than not clearing. The corollary is that genuinely bespoke hedges, which cannot be cleared, now carry a collateral cost that may exceed the value of the tailoring—which is why some hedges that would reduce a firm's risk are no longer worth doing.

Some firms were spared the clearing mandate altogether. European pension schemes were exempt from it, because clearing would have obliged them to hold cash for variation margin. The EU exemption expired in June 2023; the United Kingdom made its own permanent in 2025, so UK schemes may still keep their swaps bilateral.

The wider consequence of UMR is that initial margin has become a large, permanent, system-wide demand for high-quality assets, locked away with custodians where neither party may use it (Figure 8.1). Section 8.11 takes up that demand.

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8.5 CSA OPTIONALITY

Section 7.8 noted in passing that a CSA permitting several collateral currencies gives the poster an option. It deserves explicit treatment as a real position that must be valued and hedged.

Suppose a CSA permits cash in dollars, euros, or sterling, each earning its own overnight rate. The poster chooses which to deliver, and may switch. It will deliver whichever is cheapest to fund. Since the currencies' overnight rates and the cross-currency basis of Section 7.8 both move, the cheapest currency changes over time.

Because the collateral rate determines the discount curve, this is an option over the discount curve itself. The tools for valuing it come from two places: it is an option on the best of several rates, of the kind Section 34.10 treats, and comparing a foreign overnight rate with a domestic one requires the changes of measure of Sections 30.1 and 30.6. Neither alone is enough, which is part of why the position is so often left unpriced. Example 8.6 gives a first estimate of its size.

Example 8.6 The value of collateral choice

A dealer owes $1 billion on a portfolio under a CSA permitting dollars, euros, or sterling, so the dealer is the poster. Dollar collateral earns SOFR. To post euros instead, the dealer swaps dollars into euros: it lends dollars and receives SOFR, borrows euros and pays €STR plus the basis b (the quoting convention of Example 7.10), and earns €STR on the euros it posts. Its net return, in dollar terms, is

$$\text{SOFR} - (€STR + b) + €STR = \text{SOFR} - b$$

The euro overnight rate cancels, so what a currency's collateral earns relative to dollar collateral is its basis against the dollar with the sign reversed. With the basis at -15 basis points for euros and -10 for sterling, euro collateral earns the equivalent of SOFR plus 15 basis points, sterling SOFR plus 10, and dollars SOFR flat. The dealer posts euros, the cheapest currency to fund, and discounting at the euro collateral rate converted into dollars, rather than at SOFR, raises the effective discount rate by 15 basis points.

Take a portfolio annuity of 8, the annuity of Section 7.3: the present value of one unit a year over the portfolio's life. Posting euros rather than dollars then lowers the dealer's liability by roughly

$$\$1{,}000\text{m} \times 0.0015 \times 8 = \$12\text{m}$$

Using the annuity assumes the collateral balance stays at $1 billion over the portfolio's life, which overstates the value as a swap book amortizes.

This is only a first approximation: it prices the *current* cheapest currency, not the option to switch as relative rates move, which is worth more. Valuing it properly requires modeling the joint dynamics of the overnight rates and the cross-currency basis, a genuine multi-currency term structure problem. A dealer that ignores this and discounts on the domestic curve alone is systematically mispricing every trade under a multi-currency CSA. Two firms quoting the same trade under the same CSA can differ by millions purely on this treatment.

Similar optionality arises wherever the CSA offers a choice: eligible bonds as well as cash, a range of maturities, or substitution rights. In every case the poster will deliver the cheapest, and the receiver should price on that assumption. How much of this optionality survives depends on the agreement's vintage. The ISDA Standard CSA of 2013 was designed to remove it, and the variation margin CSAs signed since the uncleared margin rules are mostly cash-only, often in few currencies, so the option now lives mainly in legacy CSAs.

8.6 COLLATERAL TRANSFORMATION

The margin requirements of Sections 8.2 and 8.4 together create a problem for a particular kind of institution. Variation margin on cleared trades, and increasingly on bilateral ones, must be

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posted in *cash*. But the institutions with the largest derivatives hedges (pension funds, insurers, asset managers) hold almost no cash. They hold bonds, because holding cash would defeat their purpose. When margin is called they have the assets but not the form.

Collateral transformation is the business of converting bonds into cash: the fund repos its bonds, receiving cash to post as margin, and pays the repo rate while continuing to earn the bond's yield. Figure 8.2 shows the transaction, which leaves the fund with a financing leg and a rollover it did not have before. Insurers also use securities lending for the same purpose, lending corporate bonds or equities that a clearing house would not accept against government bonds or cash in a collateral upgrade trade.

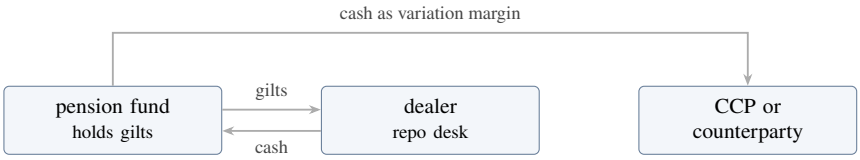


Figure 8.2: A fund repos bonds for cash to meet a margin call.

Example 8.7 The cost of transformation

A pension fund must post £500 million of cash variation margin. It holds gilts and repos them with a 2% haircut. Take illustrative rates: the repo costs 4.25%, and the cash, once posted, earns SONIA at 4.00% (the CSA rate, or price alignment interest at a clearing house).

To raise £500 million it must deliver

$$\frac{500}{0.98} = £510.2 \text{ million of gilts}$$

The gilts keep earning their yield whether or not they are repoed, so the yield drops out of the cost. The fund pays the repo rate on the cash and receives SONIA on the same cash as posted margin, a drag of 25 basis points, or

$$£500\text{m} \times 0.0025 = £1.25 \text{ million a year}$$

The haircut adds no running cost, since the fund owns the extra £10.2 million of gilts outright, but those gilts are encumbered and cannot raise cash anywhere else.

Two features make this fragile rather than merely expensive. The haircut means the fund needs *more* assets than the cash it raises, and haircuts widen under stress exactly as Example 8.3 showed: if the haircut is raised to 8% when the repo is next rolled, the same £500 million needs 500/0.92 = £543.5 million of gilts, £33.3 million more. And the transformation depends on a dealer being willing to roll the repo, a service that consumes balance sheet under the leverage ratio of Section 1.12, and that dealers withdraw when balance sheet is scarce, which is precisely when the fund needs it.

Collateral transformation therefore converts a market risk the fund has hedged into a dependence on the continued functioning of the repo market. That is a different risk, not a smaller one.

8.7 REHYPOTHECATION, SEGREGATION, AND WHAT HAPPENS ON DEFAULT

Section 8.1 distinguished title transfer from security interest. The distinction becomes concrete when the recipient of collateral fails. **Rehypothecation** is the reuse of collateral received. A dealer that receives bonds as variation margin may post them onward to meet its own obligations, and the same security can support several transactions in a chain. Both legal forms allow it: a title transfer outright, and the New York-law security interest through the right of use that Section 8.1 described. This is efficient: it means the system needs less collateral in total to

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support a given volume of activity. It is also fragile, because a chain of claims on one security unwinds messily when a link fails, and because the original poster may find that its property has traveled somewhere it cannot follow.

The protections differ sharply by margin type, and the arrangement in Figure 8.1 is what produces the difference:

- *Variation margin* is transferred outright, or under a New York-law CSA held with a right of use, and may be reused. Either way the poster has parted with control of it: if the recipient fails, the poster is an unsecured creditor for the return of equivalent assets, ranking with everyone else.
- *Initial margin* under UMR must be held by a third-party custodian and may *not* be rehypothecated. The poster remains the owner; if the recipient fails, the IM is returned rather than joining the estate.

That asymmetry is deliberate and was learned expensively. Client assets held by failed intermediaries, Lehman Brothers in 2008 and MF Global in 2011, proved far harder to recover than clients had assumed, and in several cases had been reused in ways the clients did not know about and could not have discovered. The segregation requirement for initial margin exists because the alternative was tested and failed.

Wrong-way risk in collateral. A separate hazard is taking collateral whose value falls precisely when the counterparty defaults. The extreme case, accepting a counterparty's own securities, is obvious and prohibited, but weaker versions are easy to miss: taking the bonds of a bank's sovereign as collateral against exposure to that bank, or taking energy-sector securities against exposure to an energy trading firm. In each case the collateral and the exposure share a driver, so the protection evaporates in the state where it is needed. Chapter 27 develops this as wrong-way risk in general.

8.8 DISPUTES AND RECONCILIATION

A margin call is a demand for money based on one party's valuation of a portfolio. The other party has its own valuation, and the two will differ: over the marks on illiquid trades, over which trades are in the portfolio at all, over the discount curve of Section 7.4, and over the terms of the CSA itself.

Disputes are therefore routine rather than exceptional, and the procedure for handling them is part of the arrangement. Small differences never become calls at all, because no collateral moves below the CSA's minimum transfer amount (Section 8.1). The uncleared margin rules cap it at €500,000 across initial and variation margin combined ($500,000 in the United States), which is small enough that it removes only the noise. Above it, the undisputed amount is transferred while the difference is investigated, and the CSA specifies a resolution procedure: recalculation, and failing that, polling reference dealers for quotations. Firms also reconcile portfolios on a schedule set by regulation (daily for the largest books) to catch discrepancies before they become material, and a dispute large enough and old enough must be reported to the supervisor. Three observations make this more than an operational footnote.

First, a dispute is a *funding* event. If a firm calls for $80 million and receives $50 million pending resolution, it is short $30 million while the disagreement lasts, and must fund that gap.

Second, disputes cluster exactly when they are most dangerous. Illiquid positions are hardest to mark when markets are disorderly, and both parties have an incentive to mark in their own favor. A firm relying on incoming collateral to meet its own outgoing calls has built a chain whose weakest link is a valuation disagreement, at the moment when everyone is disagreeing.

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Third, and least obviously, a dispute is expensive long after it is settled. A **netting set** (the trades with one counterparty that a single enforceable netting agreement covers) that produces repeated disputes lasting longer than the margin period of risk has that period *doubled* for the following two quarters in the dealer's counterparty capital calculation. The add-on for potential future exposure measured there (a high quantile of how large the exposure could grow, which Section 9.1 defines) scales with the square root of the period, through the maturity factor of Section 27.3, as margin does in Section 8.4. Disputing is therefore not free even for the party whose valuation is right; Example 8.8 prices the funding cost and compares the two.

Example 8.8 What a dispute costs, and when

A dealer calls a counterparty for $80 million of variation margin. The counterparty accepts $50 million and pays that, disputing the rest. The $30 million gap is outstanding for twelve days before the two sides reconcile, and the dealer funds it at 5.0%:

$$\$30\text{m} \times 0.050 \times \frac{12}{360} = \$50{,}000$$

That is the visible cost, and it is trivial. The cost that can matter more is in capital: if this is the third dispute on the netting set in two quarters to outlast the margin period of risk, the dealer's counterparty capital calculation doubles that period for the next two quarters, raising the capital held against the netting set whether or not the dealer's valuation was the correct one (Section 27.3 works through the rule).

The lesson is not that firms should concede disputes. It is that the cost of a valuation disagreement is mostly not in the disputed amount, so the investment that pays is in not disagreeing: reconciled portfolios, agreed curves, and a common margin model.

The cost of disputes also explains why the industry converged on SIMM, a choice that would otherwise look like bureaucratic uniformity. Initial margin is model-driven, and two counterparties running their own proprietary models would disagree about the number on every single call: not occasionally, but always, and by amounts far larger than the marks. A common model does not make either party's risk estimate better. It makes the two estimates the *same*, which is what converts initial margin from a standing dispute into an operational transfer. SIMM is best understood as dispute-avoidance infrastructure that happens to also measure risk.

8.9 MARGIN CALLS AS LIQUIDITY RISK

The sections above describe how collateral works. This section states the chapter's thesis, which is why any of it matters.

A hedge and the item it hedges generate cash on different schedules. The hedge is marked daily and settles daily. The hedged item (physical output to be delivered over years, a pension liability payable over decades, a bond held to maturity) does not. When the hedge loses money, the loss is a cash outflow *today*. When it gains, the offsetting loss on the hedged item is generally not a cash outflow at all. This asymmetry is invisible in a mark-to-market framework, which nets the two and reports approximately zero. It is the entire problem.

Example 8.9 The liquidity cost of a correct hedge

A power producer expects to generate 10 TWh over the next three years and hedges by selling forward at €80 per MWh. Its revenue is locked in at

$$10{,}000{,}000\ \text{MWh} \times €80 = €800\ \text{million}$$

Prices then rise to €250 per MWh. The producer's short forward position has lost

$$10{,}000{,}000 \times (€250 - €80) = €1.7\ \text{billion}$$

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and, because the position is margined, it must pay that €1.7 billion in cash, now.

Against it stands a gain of exactly €1.7 billion on the electricity it will generate. But that electricity does not exist yet. It will be produced and sold over three years, and the cash will arrive then.

The firm is better off than if it had not hedged: it has locked in €800 million of revenue in a market that might equally have fallen to €30. It is also, quite possibly, unable to pay what it owes on the day: illiquid, though not insolvent.

Set the two sides of Example 8.9 against time and they sum to approximately zero: €1.7 billion out today, about €567 million a year in over the next three years. Figure 8.3 plots the cumulative cash: the net position opens €1.7 billion short and climbs back to zero only as the last of the power is sold. The difficulty is entirely in the timing. The hedge remains economically sound right up until the firm fails. Solvency and liquidity are different questions, and derivatives collateral is the mechanism that most reliably separates them.

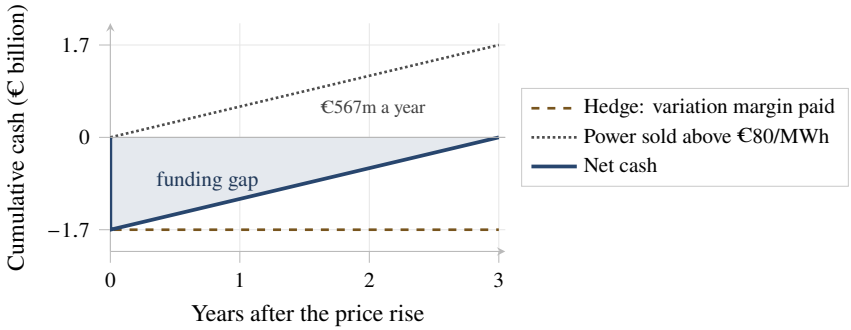


Figure 8.3: Cumulative cash in Example 8.9 (schematic: even sales, prices held at €250, no discounting).

This is why several European governments extended emergency credit lines to energy companies in 2022: the utilities in difficulty were the ones that had sold their output forward and faced Example 8.9's arithmetic at national scale, while those that had not hedged faced no margin calls and took windfall profits. The European Union's regulatory answer, in late 2022, was to widen what its clearing houses could accept: as a temporary measure, bank guarantees not backed by collateral became eligible from non-financial firms that clear directly, and public guarantees from financial and non-financial clearing members alike. The 2024 revision of the European Market Infrastructure Regulation made the relief permanent and extended it to non-financial firms that clear through a member. Section 40.12 tells the episode, and what it says about a system that prices risk correctly and liquidity not at all.

Case Study 8.1 Liability-driven investment, 2022

UK pension funds face a structural problem: liabilities extending fifty years, and not enough long-dated gilts in existence to match them. The standard solution, liability-driven investment, uses leverage (gilt repo and interest rate swaps) to obtain the required duration from a smaller pool of assets.

The leverage was collateralized, and the collateral was gilts. When the September 2022 fiscal announcement pushed gilt yields sharply higher, the funds' swap and repo positions moved against them and collateral was called. To meet the calls they sold gilts. Selling gilts pushed yields higher. Higher yields generated further calls.

The loop was not a panic: the mechanism was structural. The assets pledged as

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collateral were the same assets whose price the forced selling was driving down, so the more collateral was called, the less the remaining collateral was worth.

The Bank of England intervened by buying gilts on a temporary basis, explicitly to break the loop rather than to set a yield. The subsequent regulatory response set collateral buffers: the Bank's Financial Policy Committee (March 2023) and the Pensions Regulator (April 2023) required funds to withstand a rise of about 250 basis points in gilt yields, which caps leverage indirectly by fixing the collateral held against it. The target was liquidity rather than solvency. In January 2025 the Bank opened a contingent repo facility, activated at its discretion, through which it can lend cash against gilts to eligible insurers, defined benefit pension schemes, and liability-driven investment funds.

8.10 MEASURING AND MANAGING MARGIN LIQUIDITY

The obvious response to the previous section is to hold enough cash. The difficulty is knowing how much, since the requirement depends on a market move that has not happened. Firms therefore run **margin liquidity stress tests**, which differ from ordinary market risk measurement in three ways.

They measure gross outflow, not net loss. A value-at-risk (VaR) framework reports the net change in portfolio value. The relevant quantity here is the *cash that must leave the firm*, and the offsetting gain does not count, because it produces no cash.

They model the initial margin response, not only variation margin. A large move generates a VM call equal to the loss, *and* raises the IM requirement because volatility has risen (the procyclicality of Section 2.9), at a clearing house within days and bilaterally at the next recalibration. At a clearing house both land at once.

They assume the transformation channel narrows. A stress test that assumes normal repo access (Example 8.7) has assumed away the problem.

Example 8.10 Sizing a liquidity buffer

A fund holds a swap portfolio of \$2 billion notional with a DV01 of \$1.6 million, so a one basis point move in rates changes its value by \$1.6 million.

It stresses a 100 basis point adverse move over one week, which is severe but well within the range observed in 2022:

$$\text{VM outflow} = 100 \times \$1.6\text{m} = \$160 \text{ million}$$

Volatility rises with the move, and the initial margin model responds. The portfolio is cleared, and a clearing house's margin model reacts to a volatile week within days; SIMM, by contrast, is recalibrated only periodically. If the clearing house's initial margin on the book rises 50%, from \$60 million to \$90 million, that is a further \$30 million call.

The total requirement is \$190 million of cash within days, against a portfolio whose *value* may be unchanged once the hedged item is included. If the fund holds \$100 million of cash and expects to repo bonds for the rest, it should ask what haircut it will face at that moment, and whether dealers will still lend against its bonds at all.

The tools. Firms manage this with cash buffers, committed repo or credit lines, CSAs that permit non-cash collateral where possible, and leverage limits that keep the plausible call within reach. Each has a cost, paid continuously, to avoid an event that may not arrive. That trade-off

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is the same one Section 2.9 identified for CCP margin models, and it is resolved no more cleanly here. Sizing them requires anticipating the call, which is why the Basel Committee, CPMI, and IOSCO's final reports of January 2025 ask clearing houses to give members margin simulators that show requirements under stress scenarios, and to disclose a standardized measure of how strongly their initial margin responds to market volatility.

Why the problem is worse than each firm's own numbers suggest

Three effects make the aggregate picture worse than the sum of single-firm calculations like Example 8.10.

Calls are correlated. A large market move calls margin from everyone positioned the same way at the same moment. Each firm's stress test may be individually adequate while the system's total demand for cash exceeds what can be raised, which is precisely what turns a price move into a liquidity event.

The supply of cash contracts as demand for it rises. Repo capacity depends on dealer balance sheet (Example 8.7), so demand and supply move in opposite directions at once.

Selling to raise cash moves the price that generated the call. Where the asset sold is the asset driving the exposure (gilts sold by gilt-collateralized funds), the loop closes on itself, as the 2022 pension episode of Case Study 8.1 showed.

A single firm can estimate the second, and its own contribution to the third, as Example 8.11 does; the first, and the rest of the third, depend on what everyone else is doing.

Example 8.11 The loop, quantified

A pension fund holds £1.2 billion of gilts with a duration of 12 and closes its duration gap with £3 billion notional of twenty-year receive-fixed swaps. The twenty-year annuity at 4% is 13.590, so the swap DV01 is

$$£3\text{bn} \times 13.590 \times 0.0001 = £4.077 \text{ million per basis point}$$

Rates rise 100 basis points. On a funding-ratio basis (assets divided by liabilities, the measure a pension fund is managed to) the fund is fine: its liabilities fall by more than its assets, which is what the hedge was for. On a cash basis, assuming the swaps are cleared so that their initial margin rises with volatility (swaps margined bilaterally for variation only would call the first line alone):

Variation margin on the swaps	£407.7m
Initial margin increase as volatility doubles	£120.0m
Cash required within days	**£527.7m**

At the same instant the gilt pool has fallen 12% to £1,056m, and the repo haircut on it has widened from 2% to 8%:

	Before	After
Gilt pool	£1,200m	£1,056m
Cash raisable against it	£1,176m	£972m

Capacity fell 17.4% in the same move that raised the requirement from nothing to £527.7m. That requirement is now 54.3% of what remains. Neither number appears in a net-loss stress test, which reports approximately zero.

And the loop has not started yet. Raising £527.7m at an 8% haircut would tie up £573.6m of gilts, and on a day like this the repo may not be there. Suppose the fund sells £527.7m of gilts instead, and that selling on that scale moves twenty-year yields a further 15 basis points (modest against what was actually observed). That is another £61.2m of variation margin, met by selling another £61.2m of gilts, which moves yields 1.74 basis points more, and so on; the further fall in the gilt pool from these later rounds is ignored:

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Round	Yield move	Extra VM
1	15.00 bp	£61.2m
2	1.74 bp	£ 7.1m
3	0.20 bp	£ 0.8m

The series converges to £596.9m, a 13.1% amplification of the fund's own requirement, and it converges only because one fund's selling has a bounded price impact. The series diverges when enough funds sell at once, which is the precise sense in which this is a system-level property rather than a risk management failure: the same calculation performed by every fund individually gives the right answer for each of them and the wrong answer for all of them.

No firm-level stress test, however conservative, captures this, which is why the supervisory response, including the Financial Stability Board's work on margin preparedness, has come through financial stability channels rather than individual firm requirements.

No one behaved badly. In the energy and pension episodes of 2022 the utilities hedged as textbooks instruct, the pension funds matched their liabilities as their mandates required, and the clearing houses raised margin as their models and their regulators demanded. Every participant did what it was supposed to do, and the aggregate outcome was a liquidity crisis. Frameworks that locate such failures in imprudence will not anticipate the next one.

8.11 IS THERE ENOUGH COLLATERAL?

Clearing mandates, uncleared margin rules, and higher haircuts have each increased the high-quality assets immobilized against derivatives exposures. Is the supply adequate?

The pessimistic case is straightforward. Initial margin is segregated and cannot be reused (Section 8.2), so every dollar of IM is a dollar of government bonds or cash withdrawn from productive use for the life of a trade. Bank regulation has simultaneously increased demand for the same assets as liquidity buffers. And collateral transformation, which lets bond-holders meet cash calls, depends on dealer balance sheet that regulation has made expensive.

The optimistic case is equally straightforward. Government debt outstanding has grown enormously over the same period, so eligible collateral has expanded alongside demand. Netting and compression (tearing up offsetting trades among many participants at once) have reduced gross exposures substantially. And the pre-crisis alternative, in which large exposures went uncollateralized, was demonstrably worse.

The aggregate quantity has not been the binding constraint; the *distribution* and *timing* have. The 2022 episodes were caused not by a shortage of eligible assets but by particular firms needing cash on particular days and unable to convert what they held quickly or cheaply enough. A system can hold ample collateral in aggregate and still fail at the moment of transformation, which is why Section 8.10's stress tests focus on the channel rather than the aggregate.

8.12 WHERE THIS LEAVES VALUATION

Collateral changes what a derivative is worth, in three distinct ways that later chapters develop.

The *discount curve* is set by the CSA's collateral rate, as Section 7.4 established, and by the cheapest-to-deliver option of Section 8.5 where the CSA permits choice.

Funding enters when collateral does not fully cover the exposure. A one-way CSA, an uncollateralized trade, or a CSA with a large threshold all leave the dealer funding an exposure from its own resources. Chapter 9 prices this as a funding valuation adjustment (FVA).

Initial margin must itself be funded for the life of the trade, at a cost that no discounting convention captures. Chapter 9 prices this as MVA; Example 8.2 gave its size.

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The phrase "the value of the swap" is therefore incomplete: a swap's value depends on who holds it, under what CSA, funded how, and Chapter 9 makes that dependence explicit.

What these three do *not* capture is the proposition this chapter opened with. All of them price the *cost* of collateral: what it is worth to discount on one curve rather than another, to fund an uncollateralized exposure, to leave initial margin encumbered for ten years. None of them prices the risk of being unable to deliver collateral on the day it is called. The producer of Example 8.9 owed €1.7 billion in cash against a hedge that was working exactly as designed; no valuation adjustment would have shown a problem, because in valuation terms there was not one. That exposure is a liquidity risk, measured by the stress tests of Section 8.10 and not by any adjustment to a price. Collateral changed what derivatives are worth; it also changed what can bring a firm down, and the two are not the same list.

SUMMARY

The credit support annex sets what may be posted, at what haircut, above what threshold, and at what rate. These terms are not administrative: the collateral rate sets the discount curve, a one-way CSA is a funding commitment, and a choice of collateral currency gives the poster an option over the discount curve itself.

Variation margin settles a loss already incurred. It is netted, may be reused, and costs the poster nothing beyond the loss. Initial margin covers a possible future move. It is posted gross by both parties, segregated and barred from reuse so that it survives the failure of whoever holds it, and funded for the life of the trade, so it is a running cost. A bilateral trade immobilizes several times the collateral of a cleared one by design, which makes genuinely bespoke hedges expensive to hold.

Haircuts widen under stress, so a firm must find more assets, from a pool that has itself shrunk, to cover an exposure that has grown: the procyclicality of margin models, through a second channel. Disputes are funding events: the party that made the call funds the disputed amount until it is resolved, and disputes multiply when markets are disorderly. A firm holding bonds rather than cash depends on the repo market to transform them, and that market narrows in the same stress.

The chapter's central proposition is that a hedge and the hedged item generate cash on different schedules. A producer that sells its output forward owes margin in cash today against revenue that arrives later, so it can be better off for having hedged and still unable to pay. That is why European governments lent to energy firms in 2022, and why the UK pension episode was a collateral loop rather than a solvency problem.

Managing this requires stress tests of gross cash outflow that assume the repo channel narrows, and even those understate the problem, because calls are correlated across firms and forced sales move the prices that caused them. No participant in the energy and pension episodes of 2022 behaved imprudently, and the result was still a liquidity crisis. The system holds enough collateral in aggregate; what fails is converting it into cash on the day it is called. The valuation adjustments of Chapter 9 price the cost of collateral, not the risk of being unable to deliver it.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference for collateral, margin, and the funding that follows from them. The ISDA SIMM methodology document is the operative specification for initial margin and changes often enough that the current version is the one to use. On margin procyclicality, the Basel Committee on Banking Supervision and CPMI–IOSCO's *Review of Margining Practices* (2022) is the official record. Two final reports of January 2025 set out the policy that followed from it: the same bodies' *Transparency and*

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Responsiveness of Initial Margin in Centrally Cleared Markets: Review and Policy Proposals for clearing houses, and the Basel Committee and IOSCO's *Streamlining Variation Margin Processes and Initial Margin Responsiveness of Margin Models in Non-centrally Cleared Markets* for bilateral trades. The Financial Stability Board's *Liquidity Preparedness for Margin and Collateral Calls* (2024) is the supervisory response. The Bank of England's December 2022 *Financial Stability Report* is the place to start on liability-driven investment and gilt market functioning in the 2022 episode.

PRACTICE QUESTIONS

8.1 Explain the difference between a one-way and a two-way CSA, why a one-way CSA makes an apparently mid-market trade loss-making for the dealer, and why the rate paid on collateral is the most economically important term in the document.

8.2 A fund posts $40 million of variation margin and $25 million of initial margin, and its funding costs 60 basis points more than posted margin earns. Give the annual economic cost of each, and explain why they differ along the dimensions of what each covers, direction, and reusability.

8.3 Explain why initial margin is posted gross by both parties to a third-party custodian rather than netted between them.

8.4 A portfolio has sensitivities of $50,000 per basis point in a bucket with a risk weight of 55 bp and $90,000 per basis point in a bucket with a risk weight of 45 bp, with a prescribed correlation of 0.6. Calculate the initial margin, compare it with the undiversified sum, and explain why the result leads firms to concentrate their trades with one counterparty.

8.5 Explain why a CSA permitting several collateral currencies gives the poster an option, and why that option is an option over the discount curve.

8.6 A pension fund must post £300 million of cash margin, which earns SONIA at 4.0% once posted. It holds bonds yielding 3.8% and can repo them at 4.3% with a 3% haircut. How many bonds must it deliver, and what is the annual cost of the transformation?

8.7 An airline expects to buy 500 million gallons of jet fuel evenly over the next two years and hedges by paying a fixed $2.80 a gallon in swaps that are margined daily. Jet fuel then falls to $2.10 a gallon. Calculate the variation margin the airline must pay and the offsetting saving on its fuel bill, and state when each becomes cash. Explain why the airline can be perfectly hedged against price risk and still fail.

8.8 A fund must cover a $250 million exposure with bonds subject to a 5% haircut. How much collateral must it deliver? Recalculate at a 12% haircut, and explain why the two other things happening at the same moment make the position worse than the comparison suggests. Now suppose the $250 million is initial margin under the uncleared margin rules, posted in euro-denominated government bonds with more than five years to maturity against trades that settle in dollars. Using the standardized schedule (4% on the bonds plus the 8% currency add-on), how much must the fund deliver, and how much of that is due to the currency?

8.9 A swap has a five-day 99% risk of $20 million. Compare the collateral immobilized if it is cleared with the amount immobilized bilaterally under UMR, stating your assumptions. Why were the rules calibrated this way, and what is the cost of that design?

8.10 Explain what rehypothecation is, why it is economically efficient, and why initial margin may not be rehypothecated while variation margin may.

8.11 Give two examples of wrong-way risk in the choice of collateral, and explain the common

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feature that makes them dangerous.

8.12 A dealer calls a counterparty for $60 million of variation margin. The counterparty pays $35 million and disputes the rest, and the gap is outstanding for 20 days, which the dealer funds at 4.8%. Calculate the funding cost of the dispute. The dispute is the third on the netting set in two quarters to outlast the margin period of risk, so the period doubles from ten days to twenty for the next two quarters. If the netting set's potential future exposure add-on is $150 million and capital costs 40 basis points a year on the exposure, what does the doubling cost? Explain why a margin dispute is a funding event that clusters at the worst possible time.

8.13 A CSA gives a counterparty a threshold of $5 million and requires it to post an independent amount of $2 million. The minimum transfer amount is $500,000, and calls are rounded up to the next $100,000. The dealer's exposure is $14.23 million and it holds collateral worth $8.0 million after haircuts. Calculate the margin call. The next day the exposure is $14.6 million: is there a further call? How much of the exposure is then uncollateralized, and why?

8.14 A dealer posts $600 million of cash collateral under a CSA permitting dollars, euros, or sterling. Dollar collateral earns SOFR, the cross-currency basis against the dollar is -20 basis points for euros and -8 for sterling, and the portfolio annuity is 6. Which currency does the dealer post, and what is the approximate value of that choice relative to posting dollars? In what respect does this estimate understate the value of the CSA's optionality, and in what respect does it overstate it?

8.15 A dealer's portfolio with one counterparty holds an interest rate swap whose SIMM margin on its own is $8 million and an equity swap whose margin on its own is $5 million, and the two positions have tended to move in opposite directions. What initial margin does SIMM require on the two together? Replace the equity swap with an FX forward whose margin on its own is $5 million. SIMM prescribes a correlation of about 0.3 between the interest rate and FX margins, whatever the positions' signs: what is the margin now? Explain the difference.

FURTHER QUESTIONS

8.16 A gas producer hedges 25 TWh of output at €60 per MWh, and prices then rise to €200. What financing arrangement would have allowed the firm to meet its margin calls and survive, and what would that arrangement have cost in a year when prices did not move?

8.17 Explain the liability-driven investment loop of 2022 step by step. Identify the feature that made it self-reinforcing rather than merely costly, and explain why the regulatory response limited leverage through a minimum collateral buffer rather than by capping it directly.

8.18 A margin liquidity stress test differs from value at risk in three ways. Set them out, and explain why a firm passing a VaR-based test could still fail the liquidity one.

8.19 A $5 billion portfolio has a DV01 of $3.2 million. Stress a 150 basis point move over a week and assume the clearing house's initial margin rises 60%, from a base of $120 million. Calculate the total cash requirement, and discuss what the firm should assume about repo haircuts and dealer willingness at that moment.

8.20 Collateral was introduced to reduce credit risk, and it does. Argue that it has substituted a liquidity risk that is harder to measure and more correlated across firms than the credit risk it replaced. Then argue the other side. Which view do the 2022 episodes support?

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